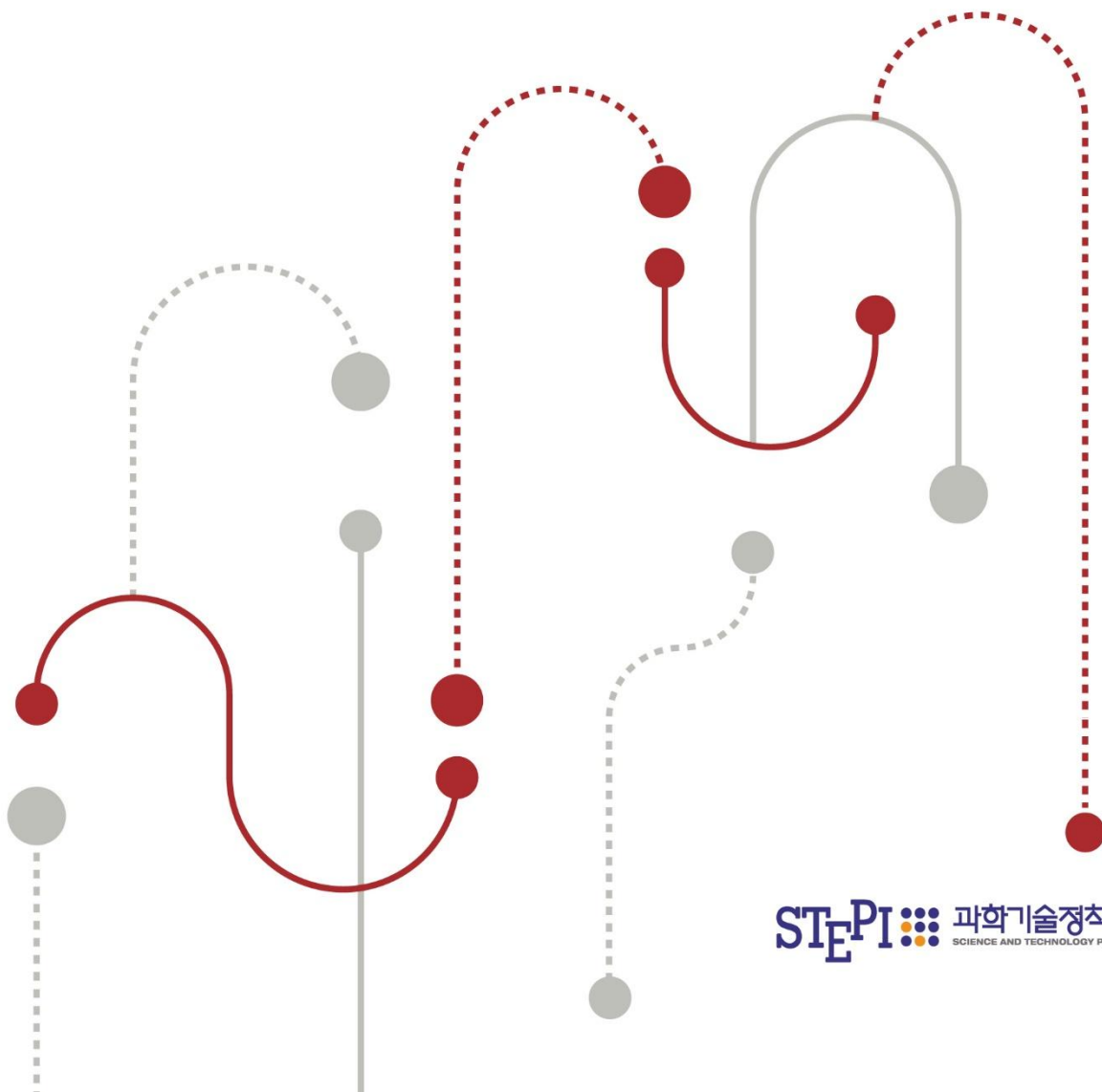


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Direction for SMEs R&D Support Based on KOSBIR Performance Analysis



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SUMMARY

Direction for SMEs R&D Support Based on KOSBIR Performance Analysis

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■ 28 Years of Support for SMEs Technology Innovation Promotion, Growing to a Similar Scale as SBIR

- The KOSBIR program, introduced in 1998 to promote SME technology development, has expanded twelvefold to ₩4.16 trillion (as of 2024), a scale comparable to that of the U.S. SBIR program.
- KOSBIR supports a much higher volume of projects across all industries (manufacturing-focused), SBIR evaluation is impact-driven (startup ratio, regional balance, job creation), whereas KOSBIR relies on quantitative indicators (sales, employment, R&D investment).

■ Significant sales and employment effects through KOSBIR support, requiring monitoring of fundamental performance indicators

- Analysis shows KOSBIR support has a positive quantitative effect on SMEs, leading to higher growth rates in sales, employment, wages, and R&D investment compared to unsupported firms.

■ KOSBIR-generated patents are quantitatively high but low in quality compared to SBIR

- KOSBIR patents are quantitatively high(28.1% of national R&D patents), significantly more than SBIR(1.53%).
- The K-PEG evaluation of Korean registered patents shows a low proportion of excellent patents and a high proportion of inadequate patents, with B1 (Normal) ranking at 33.4%

■ Direction of the SME Technology Innovation Support System

- Transition from quantity-centric to quality-centric support, emphasizing innovation and challenge by applying Government Entrepreneurship.
- Strengthen KOSBIR governance implementation and establish a data platform.

1. Advanced Analysis of SMEs Technology Innovation Support Performance

- Establish a consistent, long-term policy effectiveness analysis system by integrating environmental variables and advanced methodologies.
- Develop unified performance indicators across ministries and shift evaluations from short-term outputs to medium- and long-term market value.

2. Redefining the Target for SMEs Technology Innovation Support

- To enhance KOSBIR efficiency, clarify eligibility criteria and disclosure requirements while strengthening conditions for projects with low SME participation

- Expansion of SMEs R&D support programs by Government-Funded Research Institutes and local governments to encourage SMEs innovation attempts.

3. Pilot implementation of phased support (SBIR model)

- To enhance KOSBIR efficiency, adopt a phased linkage (SBIR) model that concentrates follow-up support on high-performing firms while offering re-challenge opportunities to others.

4. Providing incentives through performance-based follow-up support.

- To accelerate commercialization, regulatory flexibility and shared public testing infrastructure are essential for real-world validation and cost reduction.
- Establishing a differentiated incentive system for firms achieving specific commercialization milestones-such as follow-on R&D funding along with enhanced performance-based follow-up support and stronger private investment linkage.